

Problem setup:

Stream  $\{(x_1, w_1), (x_2, w_2), \dots\}$

At time  $t > k$ , where  $k$  is sample size

$P_x(x_i \text{ in Sample}) \propto w_i$  [\*]  $\rightarrow$  incorrect/incomplete

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WRS without "Replacement": [P-WRS-w/o]

Maintain a sample of size  $k$ . At each point in time  $t > k$ , the probabilities of inclusion match the following procedure

"If you construct a sample iteratively the probability that  $i$  is sampled at step  $t$  is  $\propto w_i$ "

Note that under this <sup>the</sup> marginal probability of  $x_i$  in a  $k$ -sample is NOT proportional to  $w_i$

Example  $\{w_1, w_2, w_3\}$

$k=2$

$P_r(w_i \text{ in Sample})$

$$= \frac{w_i}{w_1 + w_2 + w_3} \left[ 1 + \frac{w_j}{w_i + w_k} + \frac{w_k}{w_i + w_j} \right]$$

Thus

$P_r(w_i \text{ in Sample}) \neq w_i$

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How do we design a random  
key algorithm?

$$r_i = U[0,1]$$

$$X_i = F(r_i)$$

Sort  $X_i$  and pick top  $p-k$ .

So that  $P_r(X_i \text{ in Sample}) = P - w_{RS} - w_{LO}(i)$

How should we pick  $F$ ??

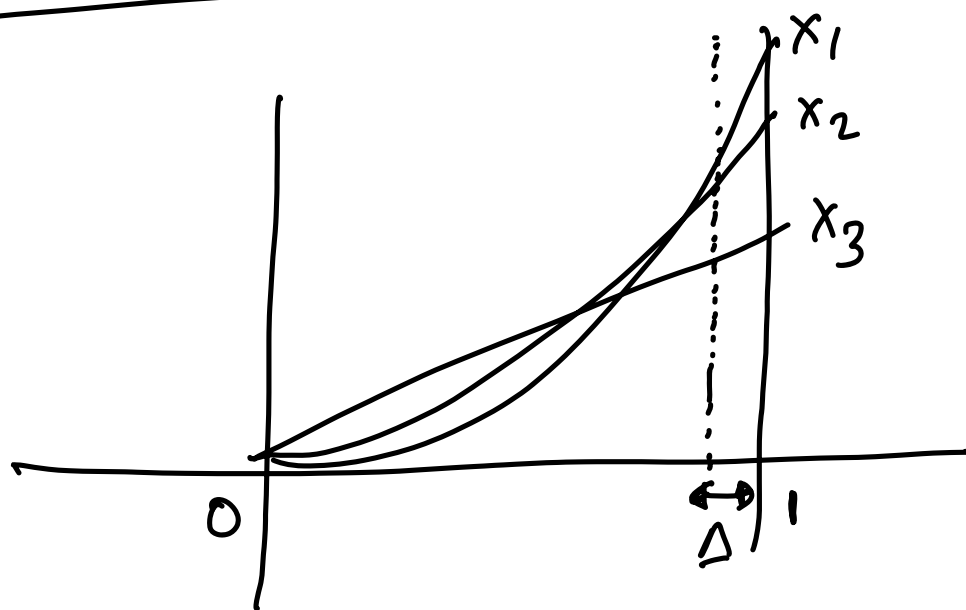
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[Intuition] How to pick  $f$

→ Simplify: let us only look at  $n=1$

→ I want probability that  $i$  lies at right most to be  $\propto w_i$

→ What should density of  $X$  look like??



If  $w_1 > w_2 > w_3$   
then  
←

$$P_n(X_i \text{ lies in } \Delta) \propto w_i$$

$$f(X_i) = f_i(x) \propto w_i \quad \underline{\text{at } 1}$$

$$f_i(x) = w g(x)$$

$$\text{where } g(1) = 1$$

$$f_i(x) = \omega g(x)$$

$$\int f_i(x) = 1 \Rightarrow \int_0^1 g(x) = \frac{1}{\omega}$$

do we know such  $g(x)$ ?

$$g(x) = [\dots]$$

$$g(x) = x^{\omega-1} \quad | \quad g(1) = 1$$

$$f_i(x) = \omega_i x^{\omega_i-1}$$

Then density should be such that

$$f_i(x) = \omega_i x^{\omega_i-1}$$

$$F_i(x) = x^{\omega_i} = \int_0^x \omega_i x^{\omega_i-1} dx$$

$$\Downarrow$$

$$X_i = r_i^{1/\omega_i}$$

$$F_i(x) = P_r(X_i \leq x)$$

$$= P_r(r_i^{1/\omega_i} \leq x)$$

$$= P_r(r_i \leq x^{\omega_i}) = x^{\omega_i}$$

Thus "Potentially"

$$f(r_i) = r_i^{1/w_i}$$

Can work.

→ we have not proved anything yet. We have just come up with "a potential" proposal that "might" work.

Proof: Full proof is out of scope.

But we can prove a lemma.

If  $r_1, r_2 \sim U(0, 1)$

$$x_1 = r_1^{1/w_1} \quad x_2 = r_2^{1/w_2}$$

$w_i > 0$

$$Pr(x_1 \leq x_2) = \frac{w_2}{w_1 + w_2}.$$

$$\begin{aligned}
& \text{Proof: } P_2(x_1 \leq x_2) \\
&= P_2(r_1^{1/\omega_1} < r_2^{1/\omega_2}) \\
&= P_2(r_1 < r_2^{\omega_1/\omega_2}) \\
&= \int_{r_2=0}^1 \int_{r_1=0}^{r_2^{\omega_1/\omega_2}} dr_1 dr_2 \\
&= \int_0^1 r_2^{\omega_1/\omega_2} dr_2 \\
&= \frac{r_2^{\omega_1/\omega_2 + 1}}{\omega_1/\omega_2 + 1} \Big|_0^1 \\
&= \frac{\omega_2}{\omega_1 + \omega_2} \quad \square
\end{aligned}$$

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Can you extend "non-random-key"  
 algo to WRS. [Exercise].